HPV Nanopore pipeline — runtime analysis workflow

INSTRUMENT · GridION / PromethION P2 Solo

R10.4.1, 400 bps · basecalling + demultiplexing (on-instrument)

output: fastq_pass/barcodeNN/*.fastq.gz (+ sequencing_summary)

1 Ingest — hpv-py

Concatenate per-barcode FASTQ into one FASTQ per sample

zcat / cat · barcode -> sample_id via sample_sheet.csv

2 Read QC — hpv-qc

Per-read length & quality stats + QC figures

NanoPlot / NanoComp · qc_report.py (pycoQC: optional)

3 Amplicon-aware length filter — hpv-qc

Length / quality window; keeps the ~1.1 kb HPV52 amp7

chopper --minlength 900 --maxlength 4800 -q 9

4 Type assignment / binning — hpv-py

Competitive map vs 12 HPV refs; best-hit/read; split per type

minimap2 -ax map-ont · hpvpipe.py bin (co-infection aware)

5 Per-type consensus — hpv-artic

ARTIC consensus/type -> consensus FASTA, VCF, trimmed BAM

artic minion · Clair3 2.x · hpvpipe.py consensus

amp3 junction run

Origin-junction handling (amp3)

Part of step 5 — the origin-spanning amplicon, recovered with two ARTIC runs:
- MAIN: 5 non-wrapping amplicons on the canonical reference
- JUNCTION: amp3 on a ROTATED ref (no negative-length crash)

rotate back + merge the seam INTO step 5's consensus => seam N = 0, continuous -> on to coverage QC

merged consensus

6 Coverage QC — hpv-py

Per-amplicon depth and dual-pool balance

pysam / mosdepth-compatible · qc_report.py

7 Summary + publication figures — hpv-py

Run summary, consensus QC and publication figure set

run_summary · consensus_qc · junction_proof · qc_report.py

results/ in the user's working directory

consensus FASTA · VCF · per-amplicon coverage

QC + publication figures · run_summary.tsv

Orchestration — the conductor model

The whole run is driven from a single orchestrator process in hpv-py (run_pipeline.py / hpvpipe.py).

Each stage's tools run in their own conda env, invoked on demand via conda run -- e.g. HPV_ARTIC_RUN='conda run -n hpv-artic'.

The envs are NOT all activated at once; the conductor calls into one env per stage.

Optional — de novo path (off step 5)

Alternative for divergent / novel / co-infection cases: flye + medaka polish. Optional — not part of every run, and does not return to the core flow.

Legend — conda environment

- hpv-py — orchestration, typing, figures
- hpv-qc — read QC, filtering, coverage
- hpv-artic — ARTIC minion + Clair3 2.x
- hpv-asm — de novo: flye + medaka

dashed outline = optional stage / branch